# Chapter 7: Blocking and Confounding in the $2^k$ Factorial Design

In many situations it is impossible to perform all of the runs in a $2^k$ factorial experiment under homogeneous conditions.

In other cases, it might be desirable to deliberately vary experimental conditions to ensure that the treatments are equally effective.

The design technique used in these situation is **blocking**. Recall that the importance of **blocking** is to **control** nuisance factor.

We will discuss blocking in a $2^k$ factorial in this chapter.

**Example 1:** An engineer is studying methods for improving the ability to detect targets on a radar scope. She considers **two factors** to be important; the amount of background noise or "**ground clutter**" on the scope and the **type of filter** placed over the screen.

An experiment is designed using **three** levels of **ground clutter** and **two filter types**. The experiment is performed by **randomly** selecting a treatment combination (ground clutter level and filter type) and then introducing a signal representing the target into the scope. The intensity of this target is increased until the operator observes it. The intensity level at detection is then measured as the response variable.

Because of **operator availability**, it is convenient to select an operator and kept him or her at the scope until all the necessary runs have been made. Furthermore, operators differ in their skill and ability to use the scope. Consequently, it seems logical to use the **operators** as **blocks**.

**Example 1 :** Four operators are **randomly** selected. Once an operator is chosen, the **order** in which the **six treatment combinations** are run is **randomly** determined. Thus, we have a **3×2 factorial experiment** run in a **randomized complete block**. The data are shown in the following table.

| Operators (Blocks) | 1 | | 2 | | 3 | | 4 | |
|---|---|---|---|---|---|---|---|---|
| Filter Type | 1 | 2 | 1 | 2 | 1 | 2 | 1 | 2 |
| Ground Clutter | | | | | | | | |
| Low | 90 | 86 | 96 | 84 | 100 | 92 | 92 | 81 |
| Medium | 102 | 87 | 106 | 90 | 105 | 97 | 96 | 80 |
| High | 114 | 93 | 112 | 91 | 108 | 95 | 98 | 83 |

```
y= c( 90, 102, 114, 86, 87, 93, 96, 106, 112, 84, 90, 91,
      100, 105, 108, 92, 97, 95, 92,  96,  98, 81, 80, 83 )

Block = factor(rep(c(1,2,3,4), each=6))
Filter = factor(rep(c(1,2), each= 3, times=4))
Clutter = factor(rep(c("Low", "Medium", "High"), times=8))
```

**Example 1:**

```
> model <- lm(y ~ (Clutter+Filter+Operator)^3)
> anova(model)
```

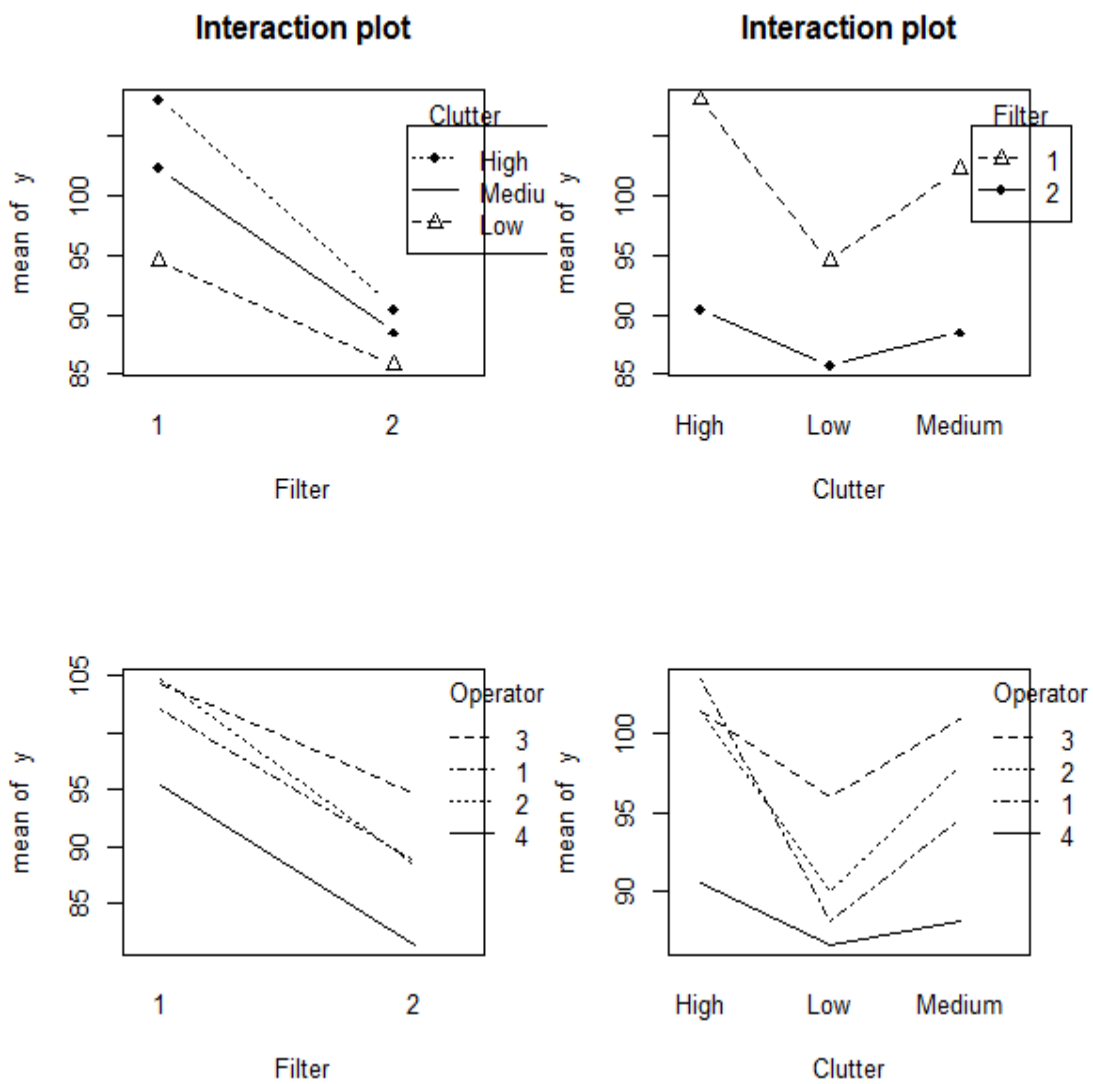
**Analysis of Variance Table**

**Response: y**

| | Df | Sum Sq | Mean Sq | F value | Pr(>F) |
|---|---|---|---|---|---|
| Clutter | 2 | 335.58 | 167.79 | | |
| Filter | 1 | 1066.67 | 1066.67 | | |
| Operator | 3 | 402.17 | 134.06 | | |
| Clutter:Filter | 2 | 77.08 | 38.54 | | |
| Clutter:Operator | 6 | 99.08 | 16.51 | | |
| Filter:Operator | 3 | 34.33 | 11.44 | | |
| Clutter:Filter:Operator | 6 | 32.92 | 5.49 | | |
| Residuals | 0 | 0.00 | | | |

# Interaction plots:

```
> par(mfrow=c(2,2))

> interaction.plot(Filter,Clutter,y,
    type="b", pch=c(18,24),
    leg.bty="o",
    main = "Interaction plot" )

interaction.plot(Clutter, Filter,y,
    type="b", pch=c(24,18),
    leg.bty="o",
    main = "Interaction plot" )

interaction.plot(Filter,Operator,y)

interaction.plot(Clutter, Operator,y)
```

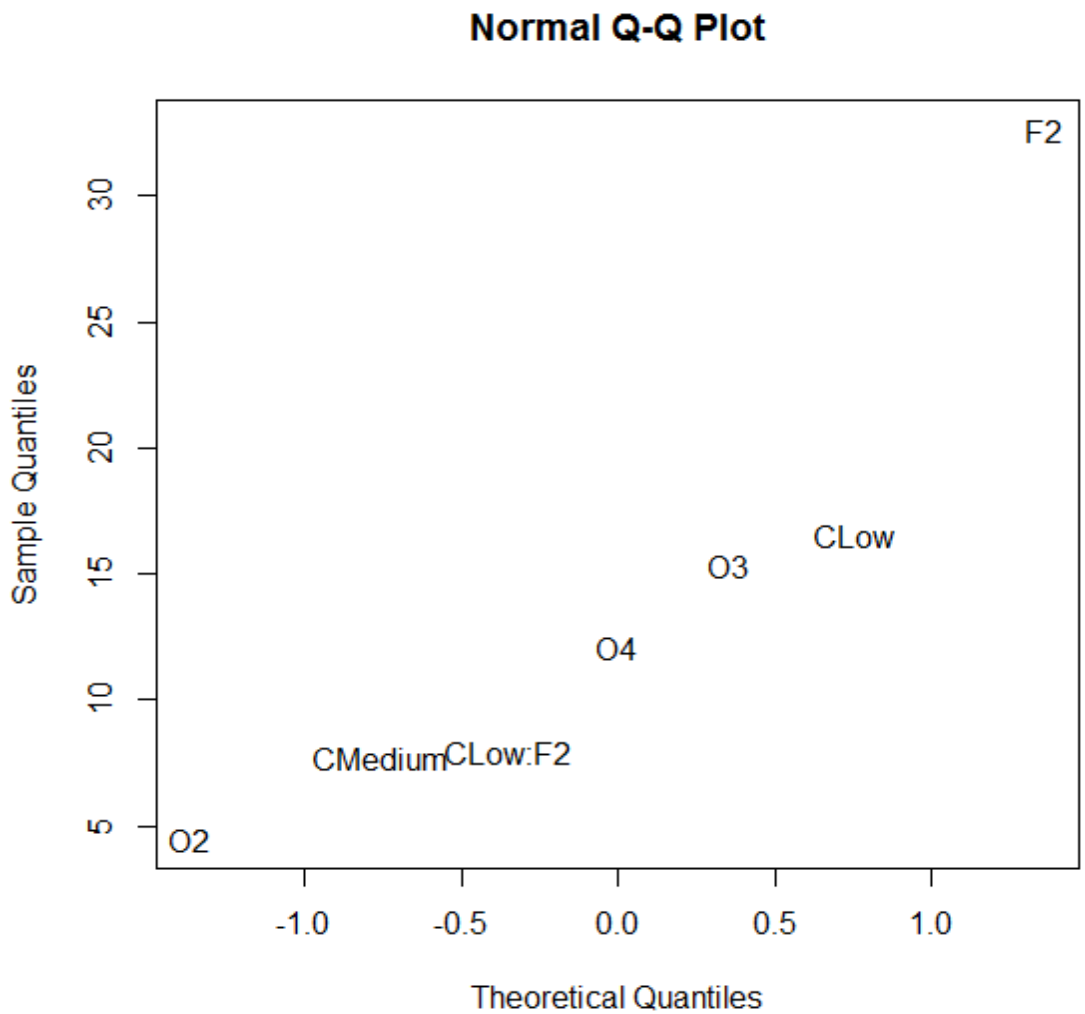


**Half-Normal Plot:** > O = Operator

> F = Filter

> C = Clutter

```
> model =
      lm(y ~(C+F+O)^3)
> model.effects =
   model$effects[2:8]
> abs.effects =
    abs(model.effects)
> qq= qqnorm(abs.effects,
      type="n")
> text(qq$x,qq$y,
    labels=names(abs.effects))
```



**Example 1:**

```
> model <- lm(y ~ Clutter+Filter+Clutter*Filter+
                  Operator+Clutter*Operator+Filter*Operator)
> anova(model)
```

**Analysis of Variance Table**

**Response: y**

| | Df | Sum Sq | Mean Sq | F value | Pr(>F) |
|---|---|---|---|---|---|
| Clutter | 2 | 335.58 | 167.79 | 30.5848 | 0.0007127 |
| Filter | 1 | 1066.67 | 1066.67 | 194.4304 | 8.479e-06 |
| Operator | 3 | 402.17 | 134.06 | 24.4354 | 0.0009200 |
| Clutter:Filter | 2 | 77.08 | 38.54 | 7.0253 | 0.0267960 |
| Clutter:Operator | 6 | 99.08 | 16.51 | 3.0101 | 0.1028509 |
| Filter:Operator | 3 | 34.33 | 11.44 | 2.0861 | 0.2035276 |
| Residuals | 6 | 32.92 | 5.49 | | |

**Example 1:**

```
> model <- lm(y ~ Clutter+Filter+Clutter*Filter+Operator)
> anova(model)
```

**Analysis of Variance Table**

**Response: y**

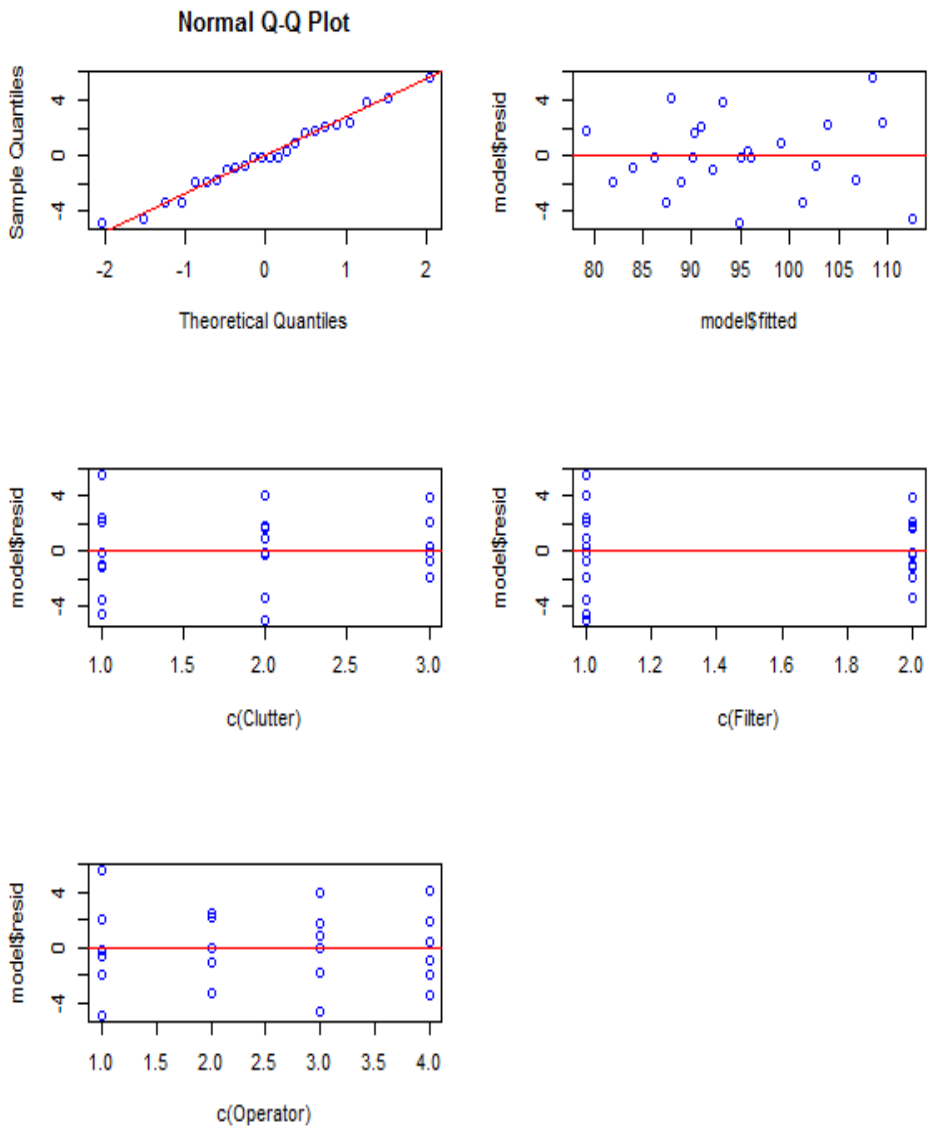
| | Df | Sum Sq | Mean Sq | F value | Pr(>F) |
|---|---|---|---|---|---|
| Clutter | 2 | 335.58 | 167.79 | 15.1315 | 0.0002527 *** |
| Filter | 1 | 1066.67 | 1066.67 | 96.1924 | 6.447e-08 *** |
| Operator | 3 | 402.17 | 134.06 | 12.0892 | 0.0002771 *** |
| Clutter:Filter | 2 | 77.08 | 38.54 | 3.4757 | 0.0575066 . |
| Residuals | 15 | 166.33 | 11.09 | | |

---

**Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1**

**Example 1:**

```
> par(mfrow=c(3,2))   > qqnorm(model$resid, col="blue")
> qqline(model$resid, col=2)
> plot(model$fitted,model$resid,
                        col="blue")
> abline(h=0, col="red")
> plot(c(Clutter), model$resid,
                        col="blue")
> abline(h=0, col="red")
> plot(c(Filter), model$resid,
                        col="blue")
>abline(h=0, col="red")
plot(c(Operator), model$resid,
                        col="blue")
abline(h=0, col="red")
```



In the case of **two randomization restrictions**, each with levels, if the number of treatment combinations in a -factor factorial design exactly equals the number of restriction levels, that is , then the **factorial design** may be run in a **Latin square**.

**Example 2:** Suppose, in example 1, **only six run can be made per day** because of the setup time required. Thus days become a second randomization restriction, resulting in the $\mathbf{6 \times 6}$ **Latin square design**, as shown in the table on the next page.

Note that the treatments A, B, C, D, E, and F represent the **six factorial treatment combinations**.

Therefore **six operators** are required now, rather than four as in the original experiment, so the number of treatment combination in the factorial design exactly equals the number of restriction levels.

So, in this design, each **operator** would be used **only** once on each day.

**Example 2:** Let use the lowercase letters $f_i$ and $g_j$ to represent the $i^{\text{th}}$ and $j^{\text{th}}$ level of filter type and ground clutter, respectively. That is $f_1 g_2$ represents filter type 1 and medium ground clutter.

The Latin square A, B, C, D, E, and F represent the factorial treatment combinations as follows:

$$A = f_1 g_1,\ B = f_1 g_2,\ C = f_1 g_3,\ D = f_2 g_1,\ E = f_2 g_2,\ \text{and } F = f_2 g_3,$$

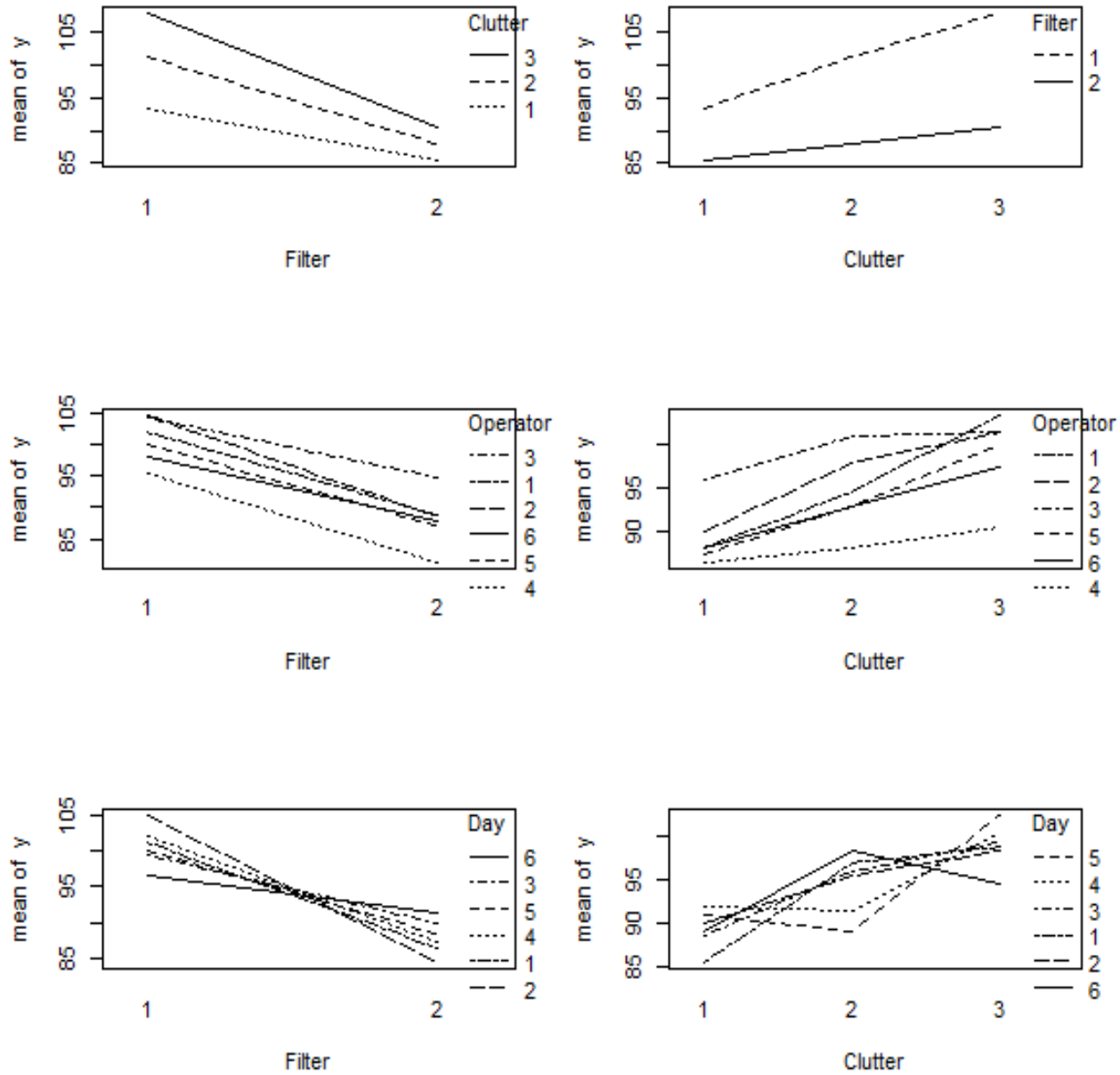
| | Operator | | | | | |
|---|---|---|---|---|---|---|
| Day | 1 | 2 | 3 | 4 | 5 | 6 |
| 1 | A=90 | B=106 | C=108 | D=81 | F=90 | E=88 |
| 2 | C=114 | A=96 | B=105 | F=83 | E=86 | D=84 |
| 3 | B=102 | E=90 | F=95 | A=92 | D=85 | C=104 |
| 4 | E=87 | D=84 | A=100 | B=96 | C=110 | F=91 |
| 5 | F=93 | C=112 | D=92 | E=80 | A=90 | B=98 |
| 6 | D=86 | F=91 | E=97 | C=98 | B=100 | A=92 |

## R Codes to Enter Data:

```
>y= c(  90, 114, 102, 87, 93, 86, 106,  96,  90, 84,112, 91,
        108, 105,  95,100, 92, 97,   81,  83,  92, 96, 80, 98,
         90, 86,  85,110, 90,100,    88, 84, 104, 91, 98, 92)
> Day = factor(rep(1:6, times=6))
> Operator = factor(rep(1:6, each=6))

>Filter = factor(c(1,1,1,2,2,2,1,1,2,2,1,2,
                  1,1,2,1,2,2,2,2,1,1,2,1,
                  2,2,2,1,1,1,2,2,1,2,1,1))

Clutter = factor(c(1,3,2,2,3,1,2,1,2,1,3,3,
                   3,2,3,1,1,2,1,3,1,2,2,3,
                   3,2,1,3,1,2,2,1,3,3,2,1))
```

# Interaction plots:

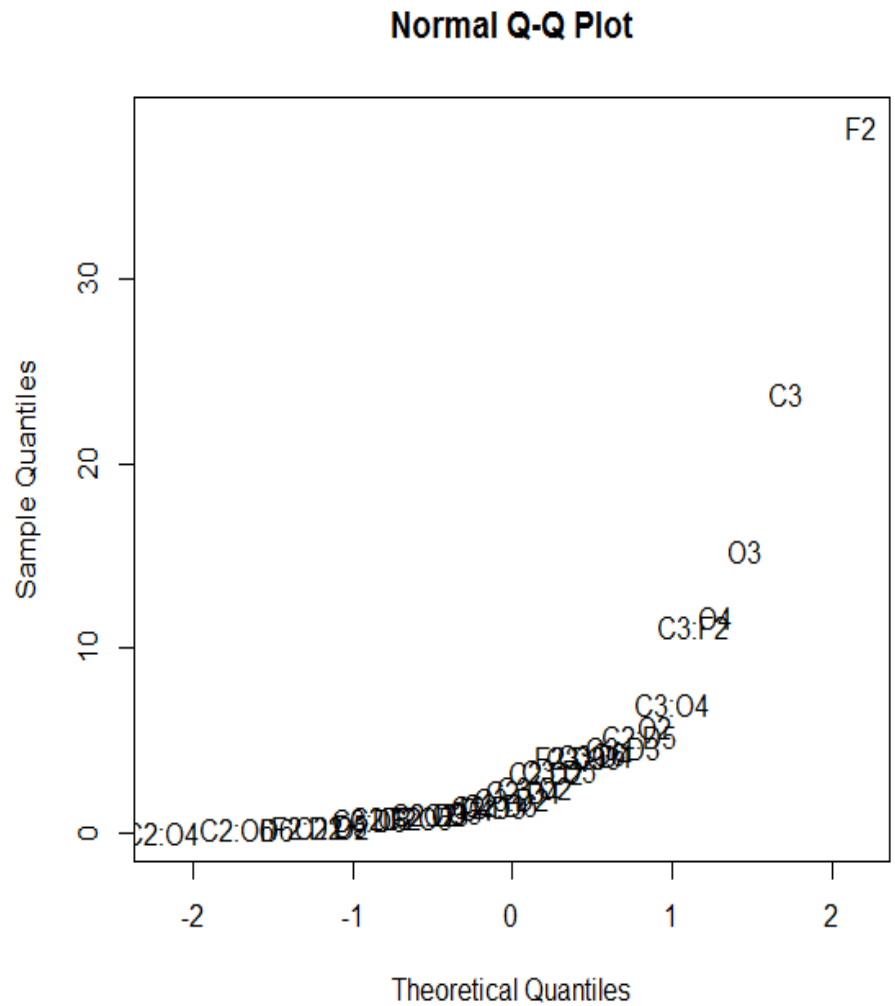
```
> par(mfrow=c(3,2))

>I nteraction.plot(Filter,
                   Clutter,y)

> interaction.plot(Clutter,
                   Filter,y)

> interaction.plot(Filter,
                   Operator,y)

> interaction.plot(Clutter,
                   Operator,y)

> interaction.plot(Filter,
                   Day,y)
> interaction.plot(Clutter,
                   Day,y)
```



**Example 2:**

>C= Clutter, F=Filter, D=Day, O= Operator

```
> model=lm(y ~ (C+F+D+O)^4)
> anova(model)

> model$effects

> model.effects =
            abs(model$effects[-1])

qq <- qqnorm(model.effects,
                        type="n")
text(qq$x, qq$y,
    labels = names(model.effects))
```



**Example 2:**

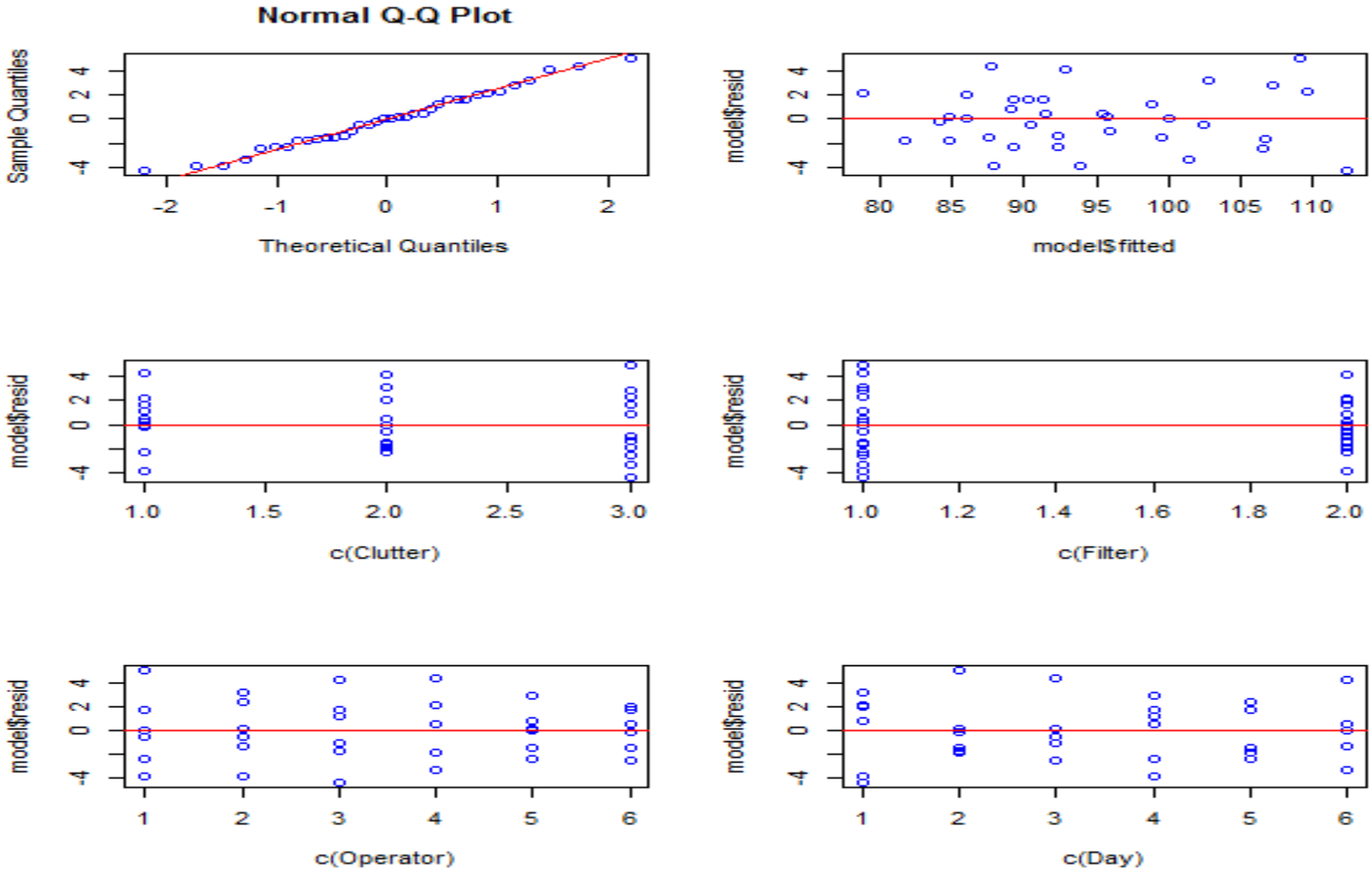
```
> model <- lm(y ~ Clutter+Filter+Clutter*Filter+Operator+Day)
> anova(model)
```

**Analysis of Variance Table**

**Response: y**

| | Df | Sum Sq | Mean Sq | F value | Pr(>F) | |
|---|---|---|---|---|---|---|
| Clutter | 2 | 571.50 | 285.75 | 28.8636 | 1.272e-06 | *** |
| Filter | 1 | 1469.44 | 1469.44 | 148.4287 | 1.040e-10 | *** |
| Operator | 5 | 428.00 | 85.60 | 8.6465 | 0.000171 | *** |
| Day | 5 | 4.33 | 0.87 | 0.0875 | 0.993365 | |
| Clutter:Filter | 2 | 126.72 | 63.36 | 6.4001 | 0.007104 | ** |
| Residuals | 20 | 198.00 | 9.90 | | | |

# Example 2:


Normal Q-Q Plot
Sample Quantiles
Theoretical Quantiles
model$resid
model$fitted
c(Clutter)
c(Filter)
c(Operator)
c(Day)

**In complete Blocks:** Consider a $2^2$ factorial design, in which only 2 runs can be made in each of 2 days (the blocks).

**Question:** Which two runs should we use in each block?

Consider the following design:

| Blocks | Treatments |
|---|---|
| 1 | (1), ab |
| 2 | a, b |

**Question:** What is the **effect of Block?**

Consider the blocks as being at a "**high**" level - Block 1 -and a "**low**" level - Block 2. Then the estimate of block effect is

$$\text{Block Effect} = \text{ average at high level - average at low level}$$

$$= \frac{ab+1}{2} - \frac{a+b}{2} = \frac{(ab-b)-(a-(1))}{2} \qquad (\text{Note: } n=1)$$

$$= \text{ Effect of A when B is high} - \text{Effect of A when B is low}$$

$$= \text{Effect of AB}$$

Consider the following design:

| Blocks | Treatments |
|---|---|
| 1 | (1), ab |
| 2 | a, b |

$$\text{Block Effect} = \text{Effect of AB}$$

We say that AB is "**confounded**" with blocks. The confounding can be seen in the $\pm$ table of signs.

| | Effect | | | | |
|---|---|---|---|---|---|
| | I | A | B | AB | Block |
| (1) | + | − | − | + | **1** |
| a | + | + | − | − | **2** |
| b | + | − | + | − | **2** |
| ab | + | + | + | + | **1** |

All runs in which **AB=+** are in **block 1** and all with **AB= −** are in **block 2**.

It is fine AB is confounded with blocks if we feel that interactions are not an issue, and don't want to estimate them.

**Question:** Consider the following design. What effect is confounded with blocks in this design?

| | Effect | | | | |
|---|---|---|---|---|---|
| | I | A | B | AB | Block |
| (1) | + | − | − | + | 1 |
| a | + | + | − | − | 2 |
| b | + | − | + | − | 1 |
| ab | + | + | + | + | 2 |

We can choose which effect is to be confounded with the two blocks.

**Question:** Consider a $2^3$ design with 4 runs in each of 2 blocks. Can we confound ABC with blocks?

For instance in a $2^3$ design, with 4 runs in each of 2 blocks, we confound ABC with blocks in the following way:

| Treatment Combinations | Factorial Effect | | | | | | | | Block |
|---|---|---|---|---|---|---|---|---|---|
| | I | A | B | AB | C | AC | BC | ABC | |
| (1) | + | − | − | + | − | + | + | − | 1 |
| a | + | + | − | − | − | − | + | + | 2 |
| b | + | − | + | − | − | + | − | + | 2 |
| ab | + | + | + | + | − | − | − | − | 1 |
| c | + | − | − | + | + | − | − | + | 2 |
| ac | + | + | − | − | + | + | − | − | 1 |
| bc | + | − | + | − | + | − | + | − | 1 |
| abc | + | + | + | + | + | + | + | + | 2 |

**Question:** Consider a $2^3$ factorial design with 2 runs in each of 4 blocks. Can we confound ABC with blocks?

For running a $2^k$ factorial design in $2,4,,8,16,\ldots$ blocks, there are useful algebraic methods to decide on a confounding scheme.

1) For 2 blocks, start with a "**defining contrast**"

$$L = \alpha_1 x_1 + \alpha_2 x_2 + \cdots + \alpha_k x_k$$

where $\boldsymbol{x_i = 1}$, if factor $i$ is **high** and $\boldsymbol{x_i = 0}$, if factor $i$ is **low** and $\boldsymbol{\alpha_i}$ is the **exponent** of **factor** $\boldsymbol{i}$ in the effect to be **confounded**.

2) Evaluate $L$ at all treatment combinations, using "**arithmetic mod 2**" where $x$(mod 2) = remainder when $x$ is divided by 2.

3) All those with $\boldsymbol{L = 0}$ **(mod 2)** go in **one block**, those with $\boldsymbol{L = 1}$ **(mod 2)** goes in **another** block.

**Example:** Consider a $\mathbf{2^3}$ factorial design in **two blocks** with $\boldsymbol{ABC} = A^1B^1C^1$ **confounded** with **blocks**.

1) We have $\alpha_1 = \alpha_2 = \alpha_3 = 1$. Therefore, we have

$$L = \alpha_1 x_1 + \alpha_2 x_2 + \alpha_3 x_3 = x_1 + x_2 + x_3$$

2) With ABC confounded with block, $L = x_1 + x_2 + x_3 (mode\ 2)$ for all treatment combinations are:

$(1): L = 0 + 0 + 0 = 0 \rightarrow \mathbf{0}\ (mode\ 2)$

$a\ : L = 1 + 0 + 0 = 1 \rightarrow 1\ (mode\ 2)$

$b\ : L = 0 + 1 + 0 = 1 \rightarrow 1\ (mode\ 2)$

$ab : L = 1 + 1 + 0 = 2 \rightarrow \mathbf{0}\ (mode\ 2)$

$c\ : L = 0 + 0 + 1 = 1 \rightarrow 1\ (mode\ 2)$

$ac :\ L = 1 + 0 + 1 = 2 \rightarrow \mathbf{0}\ (mode\ 2)$

$bc :\ L = 0 + 1 + 1 = 2 \rightarrow \mathbf{0}\ (mode\ 2)$

$abc: L = 1 + 1 + 1 = 3 \rightarrow 1\ (mode\ 2)$



3) Thus $(1), ab, ac,$ bc go to one block and $a, b, c, abc$ go to another block.

**Example:** Consider a $\mathbf{2^3}$ factorial design in **two blocks** with $\boldsymbol{AB} = A^1B^1C^0$ **confounded** with **blocks**.

1) We have $\alpha_1 = \alpha_2 = 1$ and $\alpha_3 = 0$. Thereforeherefore, we have

$$L = \alpha_1 x_1 + \alpha_2 x_2 + \alpha_3 x_3 = x_1 + x_2$$

2) With AB confounded with block, $L = x_1 + x_2\ (mode\ 2)$ for all treatment combinations are:

$(1)$: $L = 0 + 0 = 0 \rightarrow \mathbf{0}\ (mode\ 2)$

$a$ : $L = 1 + 0 = 1 \rightarrow 1\ (mode\ 2)$

$b$ : $L = 0 + 1 = 1 \rightarrow 1\ (mode\ 2)$

$ab$ : $L = 1 + 1 = 2 \rightarrow \mathbf{0}\ (mode\ 2)$

$c$ : $L = 0 + 0 = 0 \rightarrow \mathbf{0}\ (mode\ 2)$

$ac$ : $L = 1 + 0 = 1 \rightarrow 1\ (mode\ 2)$

$bc$ : $L = 0 + 1 = 1 \rightarrow 1\ (mode\ 2)$

$abc$: $L = 1 + 1 = 2 \rightarrow \mathbf{0}\ (mode\ 2)$



3) Thus $(1), ab, c,$ abc go to one block and $a, b, ac, bc$ go to another block.

For $\mathbf{2^p}$ **blocks** we would pick $\boldsymbol{p}$ **effects** to be **confounded**.

For $4 = 2^2$ blocks($4 = 2^p$ for $p = 2$) we would pick two ($p = 2$)effects to be confounded, and get two contrasts $L_1$ and $L_2$. The combinations $(L_1, L_2) = (0,0), (1,0), (0,1), (1,1)$ (mode 2) define the 4 blocks.

**Example:** Consider a $\mathbf{2^4}$ factorial design in **two blocks** with $\boldsymbol{ABCD} = A^1B^1C^1D^1$ **confounded** with **blocks**.

1) We have $\alpha_1 = \alpha_2 = \alpha_3 = \alpha_4 = 1$. Thereforeherefore, we have

$$L = \alpha_1 x_1 + \alpha_2 x_2 + \alpha_3 x_3 + \alpha_4 x_4 = x_1 + x_2 + x_3 + x_4$$

2) $L = x_1 + x_2 + x_3 + x_4$ $(mode\ 2)$ will be 0 if $\{x_1, x_2, x_3, x_4\}$ for $(1), ab, ac, ad, bc, bd, cd, abcd$ and zero otherwise.

3) Thus $(1), ab, ac, ad, bc, bd, cd, abcd$ go to one block and $a, b, c, d$, $abc, abd, acd, bcd$ go to another block.

**Example 3:** Recall that we discussed example 6.2 from textbook in last chapter.

Suppose the $2^4$ factorial for the filtration rate experiment was in **two blocks**, corresponding to different batches of formaldehyde and **ABCD** interaction was **confounded** with **blocks**.

Thus $(1), ab, ac, ad, bc, bd, cd, abcd$ go to one block and $a, b, c, d, abc, abd, acd, bcd$ go to another block.

The data have been modified by subtracting 20 from all Block 1 observations, to simulate a situation where the first batch of formaldehyde is inferior.

# A $2^4$ Factorial Design in Two Blocks:

| Run Number | Run Label | Filtration Rate (gal/h) |
|---|---|---|
| 1 | (1) | **25** |
| 2 | a | 71 |
| 3 | b | 48 |
| 4 | ab | **45** |
| 5 | c | 68 |
| 6 | ac | **40** |
| 7 | bc | **60** |
| 8 | abc | 65 |
| 9 | d | 43 |
| 10 | ad | **80** |
| 11 | bd | **25** |
| 12 | abd | 104 |
| 13 | cd | **55** |
| 14 | acd | 86 |
| 15 | bcd | 70 |
| 16 | abcd | **76** |

**R Codes to Enter Data:**

```
> y = c(25, 71, 48, 45, 68, 40, 60, 65,
        43, 80, 25, 104, 55, 86, 70, 76)
> A = rep(c(-1,1), times=8)
> B = rep(c(-1,1), each = 2, times=4)
> C = rep(c(-1,1), each = 4, times=2)
> D = rep(c(-1,1), each = 8)

> ABCD = A*B*C*D
> A = as.factor(A)
> B = as.factor(B)
> C = as.factor(C)
> D = as.factor(D)
```

Note: *ABCD* is confounded with blocks.

## R Codes to Print Data:

```
> data <- data.frame(A, B, C, D, ABCD, y)    > data
```

| | A | B | C | D | ABCD | y |
|---|---|---|---|---|---|---|
| 1 | -1 | -1 | -1 | -1 | 1 | 25 |
| 4 | 1 | 1 | -1 | -1 | 1 | 45 |
| 6 | 1 | -1 | 1 | -1 | 1 | 40 |
| 7 | -1 | 1 | 1 | -1 | 1 | 60 |
| 10 | 1 | -1 | -1 | 1 | 1 | 80 |
| 11 | -1 | 1 | -1 | 1 | 1 | 25 |
| 13 | -1 | -1 | 1 | 1 | 1 | 55 |
| 16 | 1 | 1 | 1 | 1 | 1 | 76 |

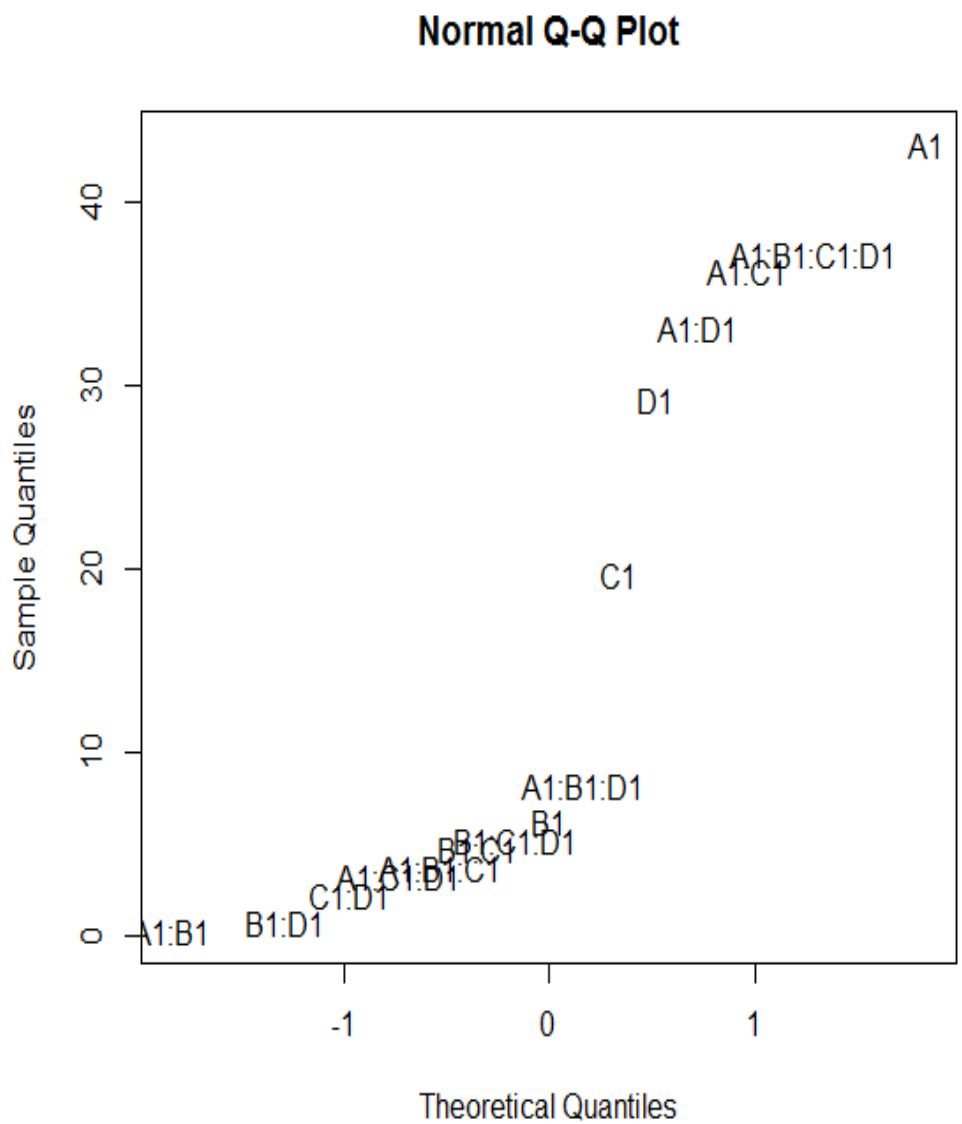
| | A | B | C | D | ABCD | y |
|---|---|---|---|---|---|---|
| 2 | 1 | -1 | -1 | -1 | -1 | 71 |
| 3 | -1 | 1 | -1 | -1 | -1 | 48 |
| 5 | -1 | -1 | 1 | -1 | -1 | 68 |
| 8 | 1 | 1 | 1 | -1 | -1 | 65 |
| 9 | -1 | -1 | -1 | 1 | -1 | 43 |
| 12 | 1 | 1 | -1 | 1 | -1 | 104 |
| 14 | 1 | -1 | 1 | 1 | -1 | 86 |
| 15 | -1 | 1 | 1 | 1 | -1 | 70 |



**Note:** One block has ABCD =1, the other ABCD = -1.

## R Codes for Half-Normal Plot:

```
> model =  lm(y ~
                (A+B+C+D)^4)
> anova(model)

> model.effects =
        abs(model$effects[-1])

> qq <- qqnorm(model.effects,
            type="n")
> text(qq$x, qq$y, labels =
          names(model.effects))
```



## R Codes for ANOVA Table:

```
>blocks <- as.factor(ABCD)
>model2= lm(y ~ A + C + D + A*C + A*D + Blocks)
> anova(model2)
```

**Analysis of Variance Table**

| | Df | Sum Sq | Mean Sq | F value | Pr(>F) |
|---|---|---|---|---|---|
| A | 1 | 1870.56 | 1870.56 | 89.757 | 5.600e-06 *** |
| C | 1 | 390.06 | 390.06 | 18.717 | 0.0019155 ** |
| D | 1 | 855.56 | 855.56 | 41.053 | 0.0001242 *** |
| Blocks | 1 | 1387.56 | 1387.56 | 66.581 | 1.889e-05 *** |
| A:C | 1 | 1314.06 | 1314.06 | 63.054 | 2.349e-05 *** |
| A:D | 1 | 1105.56 | 1105.56 | 53.049 | 4.646e-05 *** |
| Residuals | 9 | 187.56 | 20.84 | | |

For $\mathbf{2^p}$ **blocks** we would pick $\boldsymbol{p}$ **effects** to be **confounded**.

**Example 4:** Consider a $\mathbf{2^4}$ factorial design in example 3. Now, suppose we can only do **four** runs **per block**. This is $2^4$ factorial design in $4 = 2^2$ **blocks**.

Suppose we decide $\boldsymbol{ABC} = A^1B^1C^1D^0$ and $\boldsymbol{ACD} = A^1B^0C^1D^1$ are **confounded** with **blocks**.

1) We have

$$L_1 = 1.x_1 + 1.x_2 + 1.x_3 + 0.x_4 = x_1 + x_2 + x_3$$

and

$$L_2 = 1.x_1 + 0.x_2 + 1.x_3 + 1.x_4 = x_1 + x_3 + x_4$$

2) $$L_1 = 1.x_1 + 1.x_2 + 1.x_3 + 0.x_4 = x_1 + x_2 + x_3$$

$$L_2 = 1.x_1 + 0.x_2 + 1.x_3 + 1.x_4 = x_1 + x_3 + x_4$$

| | (1) | a | b | ab | c | ac | bc | abc |
|---|---|---|---|---|---|---|---|---|
| $L_1$ | 0 | 1 | 1 | 0 | 1 | 0 | 0 | 1 |
| $L_2$ | 0 | 1 | 0 | 1 | 1 | 0 | 1 | 0 |
| Block | I | IV | II | III | IV | I | III | II |

| | d | ad | bd | abd | cd | acd | bcd | abcd |
|---|---|---|---|---|---|---|---|---|
| $L_1$ | 0 | 1 | 1 | 0 | 1 | 0 | 0 | 1 |
| $L_2$ | 1 | 0 | 1 | 0 | 0 | 1 | 0 | 1 |
| Block | III | II | IV | I | II | III | I | IV |

3) **Block I** : $(1), ac, abd, bcd$

**Block II** : $b, abc, ad, cd$

**Block III**: $ab, bc, d, acd$

**Block IV**: $a, c, bd, abcd$

# A $2^4$ Factorial Design in Four Blocks:

| Run Number | Run Label | Filtration Rate (gal/h) |
|---|---|---|
| 1 | (1) | 25 |
| 2 | a | 71 |
| 3 | b | 48 |
| 4 | ab | 45 |
| 5 | c | 68 |
| 6 | ac | 40 |
| 7 | bc | 60 |
| 8 | abc | 65 |
| 9 | d | 43 |
| 10 | ad | 80 |
| 11 | bd | 25 |
| 12 | abd | 104 |
| 13 | cd | 55 |
| 14 | acd | 86 |
| 15 | bcd | 70 |
| 16 | abcd | 76 |

**R Codes to Enter Data:**

```
> y = c(25, 71, 48, 45, 68, 40, 60, 65,
        43, 80, 25, 104, 55, 86, 70, 76)
> A = rep(c(-1,1), times=8)
> B = rep(c(-1,1), each = 2, times=4)
> C = rep(c(-1,1), each = 4, times=2)
> D = rep(c(-1,1), each = 8)

> ABC = A*B*C
> ACD = A*C*D

> BD = B*D
```

Note: *ABC* and *ACD* are confounded with blocks. *BD* is also confounded with blocks.

## R Codes to Print Data:

```
>blocks = vector(length=16)

> for(i in 1:16) {
  if(ABC[i]==-1 & ACD[i]==-1) blocks[i]=1
  if(ABC[i]==1 & ACD[i]==-1) blocks[i]=2
  if(ABC[i]==-1 & ACD[i]==1) blocks[i]=3
  if(ABC[i]==1 & ACD[i]==1) blocks[i]=4
  }

> blocks = as.factor(blocks)
> data  = data.frame(A, B, C, D, blocks, ABC, ACD,BD, y)
> data
```

## R Codes to Print Data:

```
> data  = data.frame(A, B, C, D, blocks, ABC, ACD,BD, y)
> data
```

| | A | B | C | D | blocks | ABC | ACD | BD | y |
|---|---|---|---|---|---|---|---|---|---|
| 1 | -1 | -1 | -1 | -1 | 1 | -1 | -1 | 1 | 25 |
| 2 | 1 | -1 | -1 | -1 | 4 | 1 | 1 | 1 | 71 |
| 3 | -1 | 1 | -1 | -1 | 2 | 1 | -1 | -1 | 48 |
| 4 | 1 | 1 | -1 | -1 | 3 | -1 | 1 | -1 | 45 |
| 5 | -1 | -1 | 1 | -1 | 4 | 1 | 1 | 1 | 68 |
| 6 | 1 | -1 | 1 | -1 | 1 | -1 | -1 | 1 | 40 |
| 7 | -1 | 1 | 1 | -1 | 3 | -1 | 1 | -1 | 60 |
| 8 | 1 | 1 | 1 | -1 | 2 | 1 | -1 | -1 | 65 |

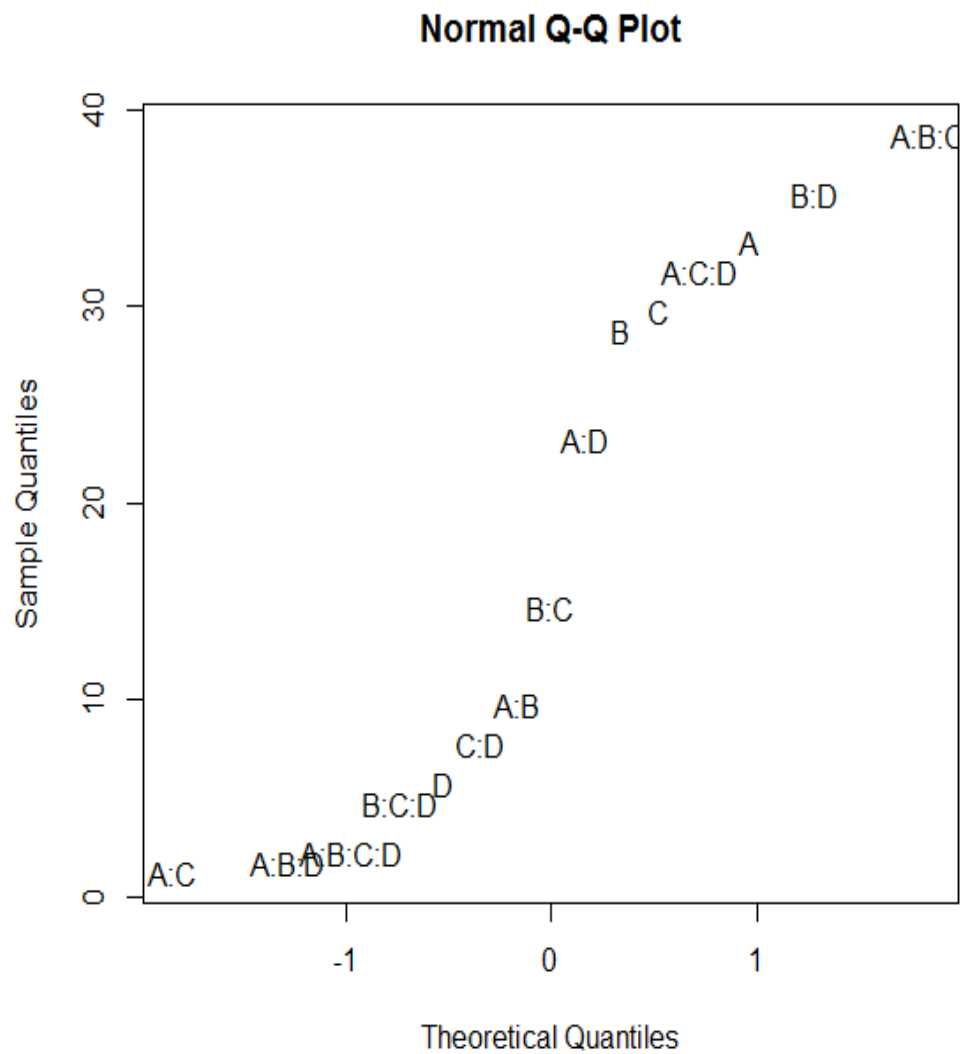
## R Codes to Print Data:

```
> data  = data.frame(A, B, C, D, blocks, ABC, ACD,BD, y)
> data
```

| | A | B | C | D | blocks | ABC | ACD | BD | y |
|---|---|---|---|---|---|---|---|---|---|
| 9 | -1 | -1 | -1 | 1 | 3 | -1 | 1 | -1 | 43 |
| 10 | 1 | -1 | -1 | 1 | 2 | 1 | -1 | -1 | 80 |
| 11 | -1 | 1 | -1 | 1 | 4 | 1 | 1 | 1 | 25 |
| 12 | 1 | 1 | -1 | 1 | 1 | -1 | -1 | 1 | 14 |
| 13 | -1 | -1 | 1 | 1 | 2 | 1 | -1 | -1 | 55 |
| 14 | 1 | -1 | 1 | 1 | 3 | -1 | 1 | -1 | 86 |
| 15 | -1 | 1 | 1 | 1 | 1 | -1 | -1 | 1 | 20 |
| 16 | 1 | 1 | 1 | 1 | 4 | 1 | 1 | 1 | 76 |

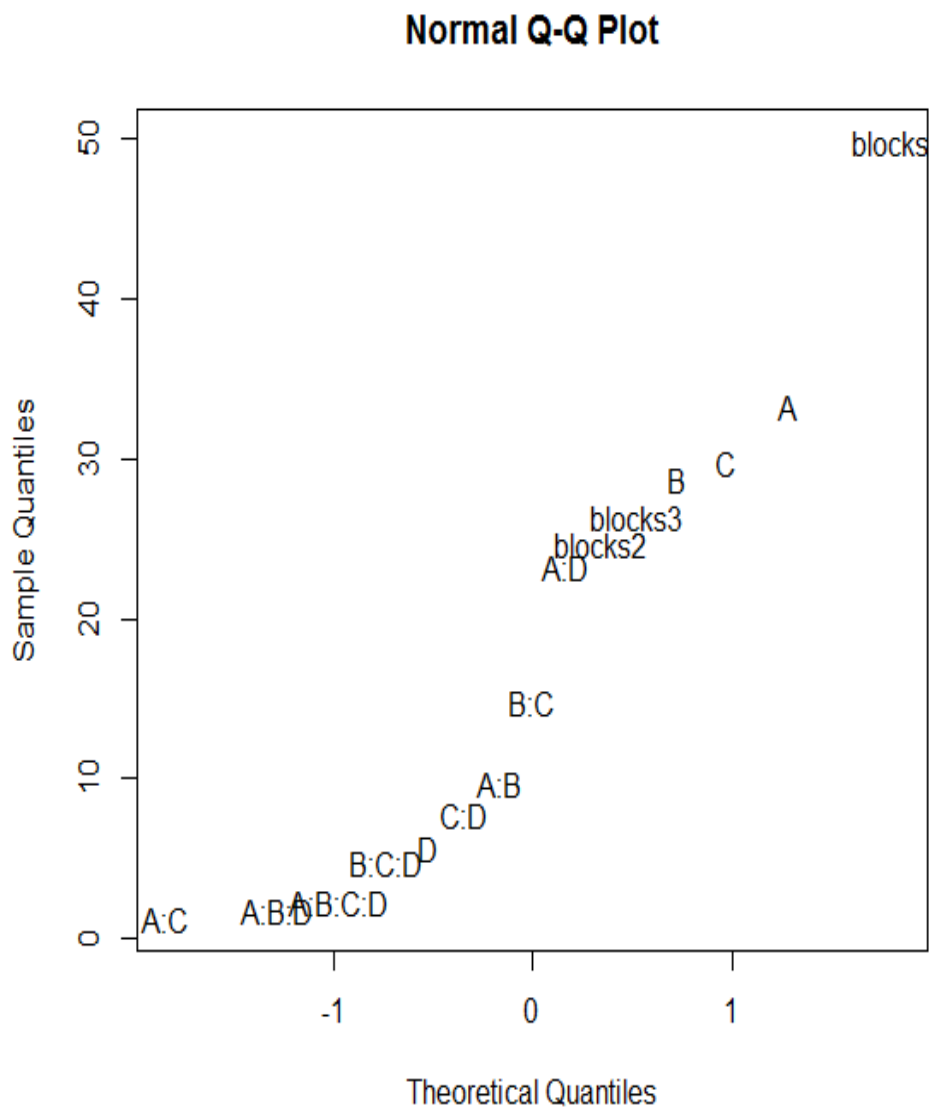
## R Codes for Half-Normal Plot:

```
> model =  lm(y ~
                (A+B+C+D)^4)
> anova(model)

> model.effects =
        abs(model$effects[-1])

> qq <- qqnorm(model.effects,
            type="n")
> text(qq$x, qq$y, labels =
          names(model.effects))
```



## R Codes for Half-Normal Plot:

```
> model1 =lm(y ~ blocks +
      A + B + C + D +
      A*B + A*C + A*D + B*C+
      C*D + A*B*D +
       B*C*D + A*B*C*D)
> anova(model1)
> model.effects =
          abs(model1$effects[-1])
> qq <- qqnorm(model.effects,
               type="n")
> text(qq$x, qq$y, labels =
           names(model.effects))
```



## R Codes for ANOVA Table:

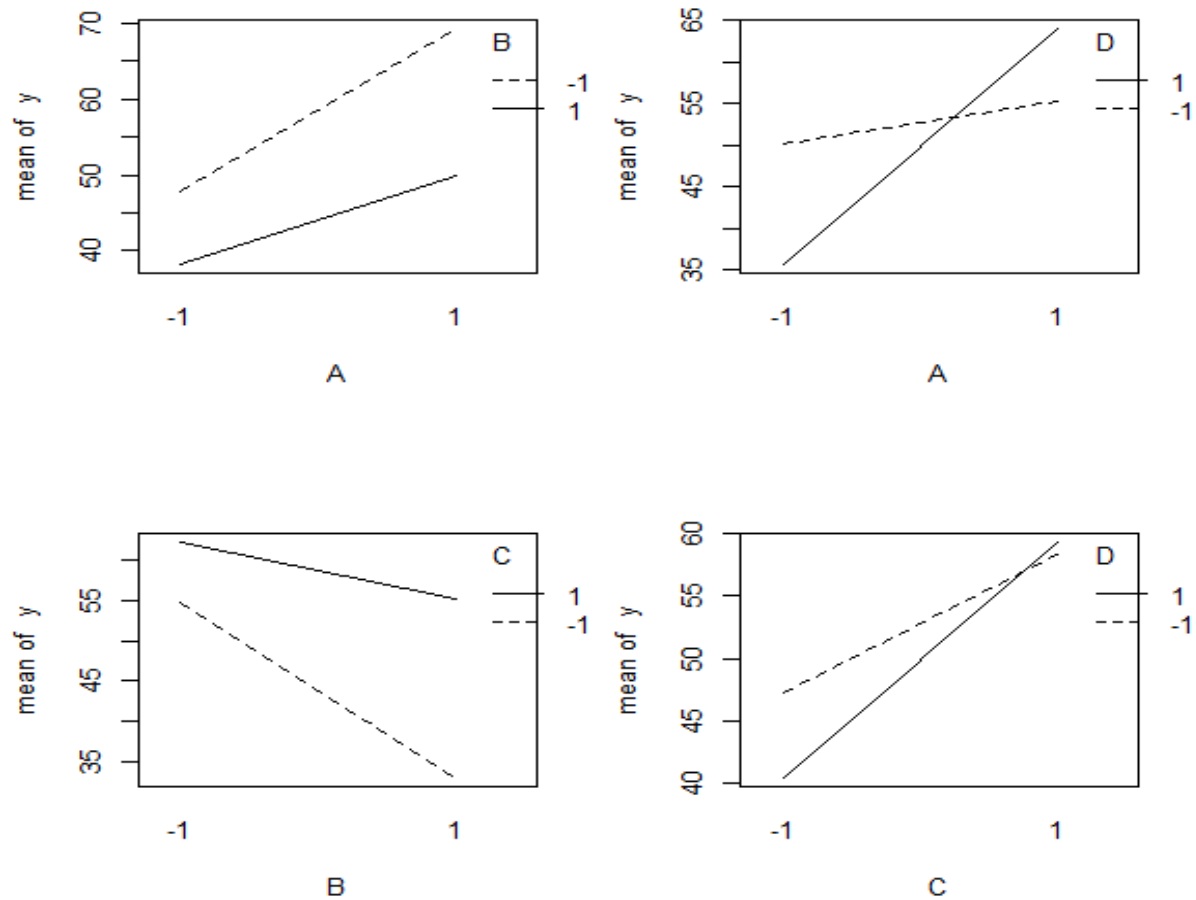
```
>model2= lm(y ~ blocks + A + B + C + D+ B*C + A*B + A*D + C*D)
> anova(model2)
```

| | Df | Sum Sq | Mean Sq | F value | Pr(>F) |
|---|---|---|---|---|---|
| blocks | 3 | 3787.7 | 1262.56 | 156.5969 | 0.0001333 *** |
| A | 1 | 1105.6 | 1105.56 | 137.1240 | 0.0003042 *** |
| B | 1 | 826.6 | 826.56 | 102.5194 | 0.0005356 *** |
| C | 1 | 885.1 | 885.06 | 109.7752 | 0.0004690 *** |
| D | 1 | 33.1 | 33.06 | 4.1008 | 0.1128484 |
| B:C | 1 | 217.6 | 217.56 | 26.9845 | 0.0065401 ** |
| A:B | 1 | 95.1 | 95.06 | 11.7907 | 0.0264444 * |
| A:D | 1 | 540.6 | 540.56 | 67.0465 | 0.0012117 ** |
| C:D | 1 | 60.1 | 60.06 | 7.4496 | 0.0524755 . |

## R Codes for Interaction Plots:

model2= lm(y ~ blocks + A + B + C + D+ B*C + A*B + A*D + C*D)

```
>par(mfrow=c(2,2))
> interaction.plot(A,B,y)
> interaction.plot(A,D,y)
> interaction.plot(B,C,y)
> interaction.plot(C,D,y)
```






The best combination seems to be A, C, D high and B low.

For $2^p$ **blocks** we would pick $\boldsymbol{p}$ **effects** to be **confounded**.

If **two** effects are **confounded** with **blocks**, then so is **their product**, which is defined by "**multiplication mod 2**": $A^0 = A^2 = 1, A^1 = A$

If $\boldsymbol{AB}$ and $\boldsymbol{AC}$ effects are to be confounded with blocks, then $AB * AC = A^2BC = \boldsymbol{BC}$ is confounded too.

If $\boldsymbol{ABC}$ and $\boldsymbol{ACD}$ effects are to be confounded with blocks, then $ABC * ACD = A^2BC^2D = \boldsymbol{BD}$ is confounded too.

If $\boldsymbol{ABC}$ and $\boldsymbol{ABCD}$ effects are to be confounded with blocks, then $ABC * ABCD = A^2B^2C^2D = \boldsymbol{D}$ is confounded too.

The choice of $\boldsymbol{p}$ **effects** to be **confounded** with blocks is critical.

Table 7.9 of textbook on page of **322** present a list of useful designs.

The function **FrF2** function in the R **package FrF2** can find a $2^k$ **factorial design** in **block** of size $2^q$.

As an example consider confounding a $2^4$ factorial design in blocks of size $4 = 2^2$. Suppose **ABD** and **BCD** should be confounded with blocks.

The following R codes generate this design.

```
> library(FrF2)
> design = FrF2(16,4, blocks=c("ABC", "BCD"),
            alias.block.2fis=TRUE, randomize=FALSE)
```

In practice the treatment combinations within each block would be randomized to experimental units in that block by removing the option **randomize = FALSE**.

## $2^4$ factorial design in blocks of size $4 = 2^2$:

> design

| | run.no | run.no.std.rp | Blocks | A | B | C | D |
|---|---|---|---|---|---|---|---|
| 1 | 1 | 1.1.1 | 1 | -1 | -1 | -1 | -1 |
| 2 | 2 | 7.1.2 | 1 | -1 | 1 | 1 | -1 |
| 3 | 3 | 12.1.3 | 1 | 1 | -1 | 1 | 1 |
| 4 | 4 | 14.1.4 | 1 | 1 | 1 | -1 | 1 |
| | run.no | run.no.std.rp | Blocks | A | B | C | D |
| 5 | 5 | 2.2.1 | 2 | -1 | -1 | -1 | 1 |
| 6 | 6 | 8.2.2 | 2 | -1 | 1 | 1 | 1 |
| 7 | 7 | 11.2.3 | 2 | 1 | -1 | 1 | -1 |
| 8 | 8 | 13.2.4 | 2 | 1 | 1 | -1 | -1 |
| … | …. | …….. | ….. | … | … | … | …. |

**class=design, type= FrF2.blocked**

**NOTE: columns run.no and run.no.std.rp are annotation, not part of the data frame**

The choice of $p$ **effects** to be **confounded** with blocks is critical.

Table 7.9 of textbook on page of 316 present a list of useful designs.

Box et al. (Box, G.E. P., Hunter, W.G., and Hunter, J.S. (1978), "*Statistics for Experimenters*", John Wiley & Sons, New York, first edition) also provide tables for block defining contrasts that will result in a **minimum number of low-order interactions** being **confounded** with **blocks** in a $\mathbf{2^k}$ blocked design. However, Sun et al. (Sun, D. X., Wu, C.F.J., and Chen, Y. Y. (1997), "*Optimal blocking schemes for* $2^p$ *and* $2^{n-p}$ *designs*", *Technometrics*, **39**, 298-307. ) provide an extensive catalog of block defining contrast for $2^k$ designs that will result in best designs with regard to one of several quality criteria such as estimability order.

When **not** specified by the user, the function **FrF2** in the R package FrF2 uses the block defining contrast from ***Sun et al.***'s.

**Partial Confounding:** To have an estimate of error, we have to either drop certain effects from the model, or replicate the design.

If we replicate, we can either have "**complete**" confounding or "**partial**" confounding.

**Complete confounding** is when we confound the same effects with blocks in each replication.

**Partial confounding** is when we confound different effects with blocks in each replication.

**Partial confounding** is usually **better**, since we then get **estimates** of **effects** from the replications in which they are not confounded.

**Example 5 (Problem 7.24 from Textbook):** An Engineer is interested in the effects of cutting speed (A), tool geometry (B), and angle (C) on the life (in hours) of a machine tool. Two levels of each factor are chosen and three replicates of a $2^3$ factorial design are run. The results are as follow:

| A | B | C | Treatment Combination | Replicate I | Replicate II | Replicate III |
|---|---|---|---|---|---|---|
| - | - | - | (1) | 22 | 31 | 25 |
| + | - | - | a | 32 | 43 | 29 |
| - | + | - | b | 35 | 34 | 50 |
| + | + | - | ab | 55 | 47 | 46 |
| - | - | + | c | 44 | 45 | 38 |
| + | - | + | ac | 40 | 37 | 36 |
| - | + | + | bc | 60 | 50 | 54 |
| + | + | + | abc | 39 | 41 | 47 |

Suppose that ABC, AB and BC were confounded in replicate I, II and III, respectively.

**Example 5:** **Three replicates** of a $2^3$ factorial design are run, in **two blocks** each.

Replicate 1: **ABC** was confounded with blocks. Therefore, $L = x_1 + x_2 + x_3 = 0$ for $(1), ab, ac, bc$ and $L = 1$ for $a, b, c, abc$.

Replicate 2: **AB** was confounded with blocks. Therefore, $L = x_1 + x_2 = 0$ for $(1), ab, abc, c$ and $L = 1$ for $a, b, ac, bc$.

Replicate 2: **BC** was confounded with blocks. Therefore, $L = x_2 + x_3 = 0$ for $(1), a, bc, abc$ and $L = 1$ for $ab, b, c, ac$.

Although there are confounding, we should be able to estimate all effects including interaction effects.

## R Codes to Enter Data:

| Rep 1 | | Rep 2 | | Rep 3 | |
|---|---|---|---|---|---|
| Block 1 | Block 2 | Block 3 | Block 4 | Block 5 | Block 6 |
| (1)=22 | a =32 | (1)=31 | a=43 | (1)=25 | b=50 |
| ab=55 | b=35 | c=45 | b=34 | a=29 | c=38 |
| ac=40 | c=44 | ab=47 | ac=37 | bc=54 | ab=46 |
| bc=60 | abc=39 | abc=41 | bc=50 | abc=47 | ac=36 |

```
> y = c(22, 55, 40, 60, 32, 35, 44, 39,
        31, 45, 47, 41, 43, 34, 37, 50,
        25, 29, 54, 47, 50, 38, 46, 36)

> A = c(-1,1,1,-1,1,-1,-1,1,-1,-1,1,1,1,-1,1,-1,-1,1,-1,1,-1,-1,1,1)

> B =c(-1,1,-1,1,-1,1,-1,1, -1,-1,1,1,-1,1,-1,1,-1,-1,1,1,1,-1,1,-1)

> C =c(-1,-1,1,1,-1,-1,1,1,-1,1,-1,1,-1,-1,1,1,-1,-1,1,1,-1,1,-1,1)
```

## R Codes to Define Factors and Print Data:

```
> ABC = A*B*C
> AB  = A*B
> BC  = B*C

> Rep = as.factor(rep(c("I", "II", "III"), each = 8))
> Block = as.factor(rep(1:6, each=4))

> A = as.factor(A)
> B = as.factor(B)
> C = as.factor(C)

> data = data.frame(A, B, C, Rep, Block, ABC, AB,BC,  y)
> data
```

## Data:

| | A | B | C | Rep | Block | ABC | AB | BC | y |
|---|---|---|---|---|---|---|---|---|---|
| 1 | -1 | -1 | -1 | I | 1 | -1 | 1 | 1 | 22 |
| 2 | 1 | 1 | -1 | I | 1 | -1 | 1 | -1 | 55 |
| 3 | 1 | -1 | 1 | I | 1 | -1 | -1 | -1 | 40 |
| 4 | -1 | 1 | 1 | I | 1 | -1 | -1 | 1 | 60 |
| 5 | 1 | -1 | -1 | I | 2 | 1 | -1 | 1 | 32 |
| 6 | -1 | 1 | -1 | I | 2 | 1 | -1 | -1 | 35 |
| 7 | -1 | -1 | 1 | I | 2 | 1 | 1 | -1 | 44 |
| 8 | 1 | 1 | 1 | I | 2 | 1 | 1 | 1 | 39 |
| 9 | -1 | -1 | -1 | II | 3 | -1 | 1 | 1 | 31 |
| 10 | -1 | -1 | 1 | II | 3 | 1 | 1 | -1 | 45 |
| 11 | 1 | 1 | -1 | II | 3 | -1 | 1 | -1 | 47 |
| 12 | 1 | 1 | 1 | II | 3 | 1 | 1 | 1 | 41 |
| 13 | 1 | -1 | -1 | II | 4 | 1 | -1 | 1 | 43 |
| 14 | -1 | 1 | -1 | II | 4 | 1 | -1 | -1 | 34 |
| 15 | 1 | -1 | 1 | II | 4 | -1 | -1 | -1 | 37 |
| 16 | -1 | 1 | 1 | II | 4 | -1 | -1 | 1 | 50 |
| 17 | -1 | -1 | -1 | III | 5 | -1 | 1 | 1 | 25 |
| 18 | 1 | -1 | -1 | III | 5 | 1 | -1 | 1 | 29 |
| 19 | -1 | 1 | 1 | III | 5 | -1 | -1 | 1 | 54 |
| 20 | 1 | 1 | 1 | III | 5 | 1 | 1 | 1 | 47 |
| 21 | -1 | 1 | -1 | III | 6 | 1 | -1 | -1 | 50 |
| 22 | -1 | -1 | 1 | III | 6 | 1 | 1 | -1 | 38 |
| 23 | 1 | 1 | -1 | III | 6 | -1 | 1 | -1 | 46 |
| 24 | 1 | -1 | 1 | III | 6 | -1 | -1 | -1 | 36 |

When the levels of one factor (**Blocks**) make sense only **within** the levels of another factor (**Replicates**) we say that the first is "**nested**" within the second.

A way to indicate this in R is as:

```
> model = lm(y ~ Rep + Block%in%Rep
                + A + B + C + A*B + A*C + B*C + A*B*C)
```

**Note:** SS for blocks are calculated within each replicate since block make sense only within replicates. Let $\bar{y}_{ij.}$ be the average of responses in block $j$ of replicate $i$ and $\bar{y}_{i..}$ be the overall average of responses of replicate $i$. Then

$$SS_{Block(Rep)} = 4\sum_{i=1}^{3}\sum_{j=1}^{6}(\bar{y}_{ij.} - \bar{y}_{i..})^2$$

$$= SS_{ABC\ in\ Rep\ I} + SS_{AB\ in\ Rep\ II} + SS_{BC\ in\ Rep\ III}$$

| Rep 1 | | Rep 2 | | Rep 3 | |
|---|---|---|---|---|---|
| Block 1 | Block 2 | Block 3 | Block 4 | Block 5 | Block 6 |
| (1)=22 | a =32 | (1)=31 | a=43 | (1)=25 | b=50 |
| ab=55 | b=35 | c=45 | b=34 | a=29 | c=38 |
| ac=40 | c=44 | ab=47 | ac=37 | bc=54 | ab=46 |
| bc=60 | abc=39 | abc=41 | bc=50 | abc=47 | ac=36 |

$$\bar{y}_{11.} = \frac{22+55+40+60}{4} = 44.25, \bar{y}_{12.} = \frac{32+35+44+39}{4} = 37.50$$

$$\bar{y}_{1..} = \frac{22+55+40+60+32+35+44+39}{8} = 40.875$$

Similarly, we have

$$\bar{y}_{23.} = 41.00, \quad \bar{y}_{24.} = 41.00, \quad \bar{y}_{2..} = 41.00$$

$$\bar{y}_{35.} = 38.75, \quad \bar{y}_{36.} = 42.50 \quad \bar{y}_{3..} = 40.625$$

$$SS_{Block(Rep)} = 4\sum_{i=1}^{3}\sum_{j=1}^{6}(\bar{y}_{ij.} - \bar{y}_{i..})^2$$

$$= 4\left[\begin{array}{c}(44.25-40.875)^2+(37.50-40.875)^2+\cdots \\ +(42.50-40.625)^2\end{array}\right] = 119.25$$

$$SS_{Block(Rep)} = 4\sum_{i=1}^{3}\sum_{j=1}^{6}(\bar{y}_{ij.} - \bar{y}_{i..})^2$$

**R Codes to Calculate SS for Blocks:**

```
> block.means <- vector(length=6)
> for(j in 1:6) block.means[j] <- mean(y[Block ==j])
> rep.means <- rep(c(mean(y[Rep == "I"]),mean(y[Rep == "II"]),
                                        mean(y[Rep == "III"])), each=2)
> cbind(block.means,rep.means)

> SS.Blocks.in.Rep <- 4*sum((block.means-rep.means)^2)
> SS.Blocks.in.Rep
```

**[1] 119.25**

| Rep 1 | | Rep 2 | | Rep 3 | |
|---|---|---|---|---|---|
| Block 1 | Block 2 | Block 3 | Block 4 | Block 5 | Block 6 |
| (1)=22 | a =32 | (1)=31 | a=43 | (1)=25 | b=50 |
| ab=55 | b=35 | c=45 | b=34 | a=29 | c=38 |
| ac=40 | c=44 | ab=47 | ac=37 | bc=54 | ab=46 |
| bc=60 | abc=39 | abc=41 | bc=50 | abc=47 | ac=36 |

$$SS_{ABC\ in\ Rep\ I} = \frac{[(32+35+44+39)-(22+55+40+60)]^2}{8}$$
$$= 91.125$$

$$SS_{AB\ in\ Rep\ II} = \frac{[(43+34+37+50)-(31+45+47+41)]^2}{8}$$
$$= 0.000$$

$$SS_{BC\ in\ Rep\ III} = \frac{[(50+38+46+36)-(25+29+54+47)]^2}{8}$$
$$= 28.125$$

$$SS_{Block(Rep)} = SS_{ABC\ in\ Rep\ I} + SS_{AB\ in\ Rep\ II} + SS_{BC\ in\ Rep\ III}$$
$$= 91.125 + 0.000 + 28.125 = 119.125$$

$$SS_{Block(Rep)} = SS_{ABC\ in\ Rep\ I} + SS_{AB\ in\ Rep\ II} + SS_{BC\ in\ Rep\ III}$$

**R Codes to Calculate SS for Blocks:**

```
> SSABC.confounded =((sum(y[Rep=="I" & ABC==1])
                        -sum(y[Rep=="I" & ABC==-1]))^2)/8
> SSAB.confounded  = ((sum(y[Rep=="II" & AB==1])
                        -sum(y[Rep=="II" & AB==-1]))^2)/8
> SSBC.confounded  = ((sum(y[Rep=="III" & BC==1])
                        -sum(y[Rep=="III" & BC==-1]))^2)/8

> SS.Blocks.in.Rep = SSABC.confounded  +
                     SSAB.confounded  + SSBC.confounded

> SS.Blocks.in.Rep
[1] 119.25
```

**ANOVA table:**

> anova(model)

**Analysis of Variance Table**

**Response: y**

| | Df | Sum Sq | Mean Sq | F value | Pr(>F) | |
|---|---|---|---|---|---|---|
| Rep | 2 | 0.58 | 0.29 | 0.0079 | 0.9921768 | |
| A | 1 | 0.67 | 0.67 | 0.0180 | 0.8957978 | |
| B | 1 | 770.67 | 770.67 | 20.7672 | 0.0008205 | *** |
| C | 1 | 280.17 | 280.17 | 7.5497 | 0.0189697 | * |
| Rep:Block | 3 | 119.25 | 39.75 | 1.0711 | 0.4009691 | |
| A:B | 1 | 25.00 | 25.00 | 0.6737 | 0.4292019 | |
| A:C | 1 | 468.17 | 468.17 | 12.6157 | 0.0045372 | ** |
| B:C | 1 | 22.56 | 22.56 | 0.6080 | 0.4519957 | |
| A:B:C | 1 | 0.06 | 0.06 | 0.0017 | 0.9680003 | |
| Residuals | 11 | 408.21 | 37.11 | | | |

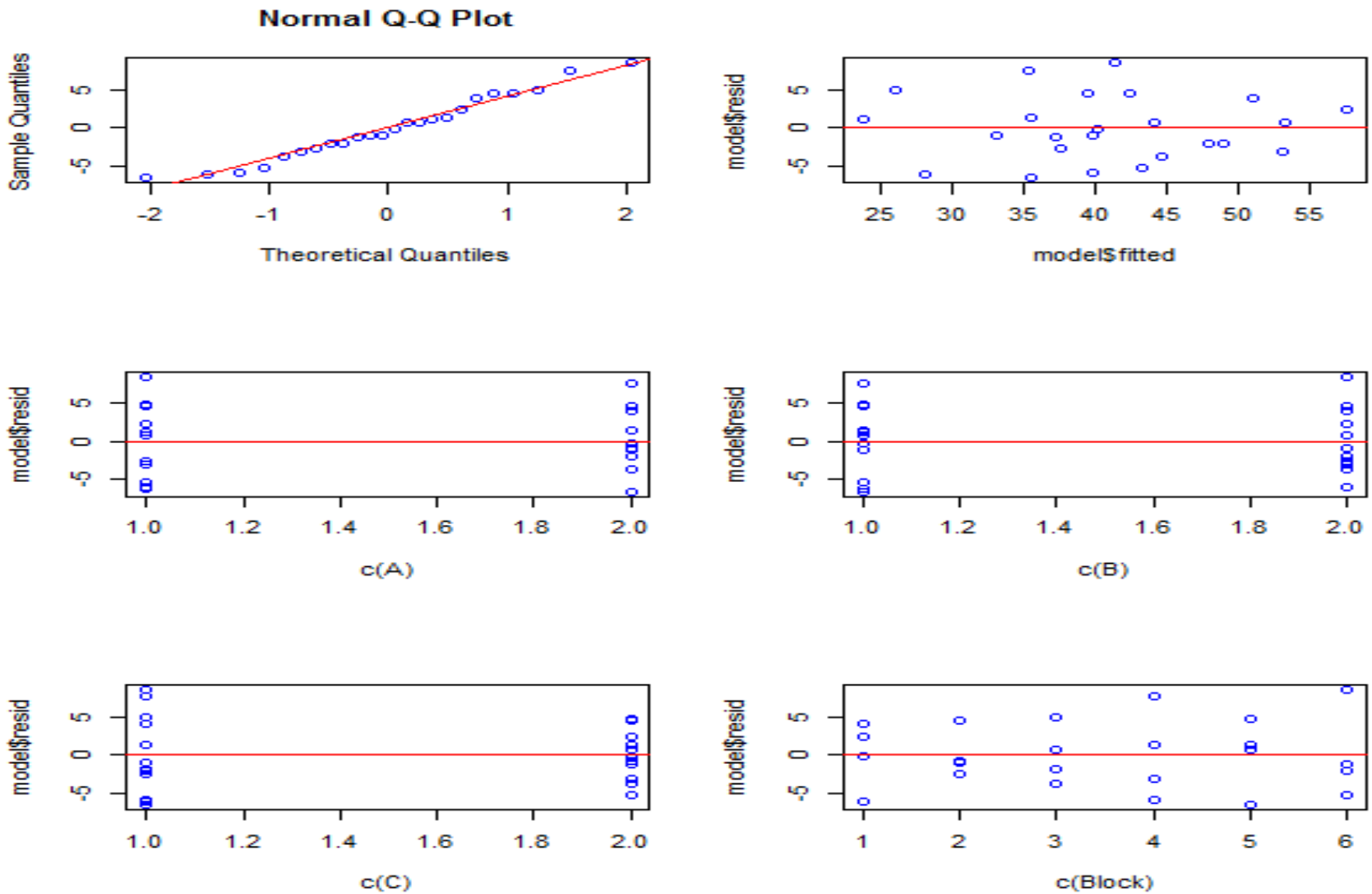
**ANOVA table:**

```
> model2 <- lm(y ~ Rep + Block%in%Rep + A + B+ C + A*C)
> anova(model2)
```

**Analysis of Variance Table**

**Response: y**

| | Df | Sum Sq | Mean Sq | F value | Pr(>F) |
|---|---|---|---|---|---|
| Rep | 2 | 0.58 | 0.29 | 0.0090 | 0.9910877 |
| A | 1 | 0.67 | 0.67 | 0.0205 | 0.8882566 |
| B | 1 | 770.67 | 770.67 | 23.6695 | 0.0002501 *** |
| C | 1 | 280.17 | 280.17 | 8.6048 | 0.0108984 * |
| Rep:Block | 3 | 119.25 | 39.75 | 1.2208 | 0.3388282 |
| A:C | 1 | 468.17 | 468.17 | 14.3788 | 0.0019821 ** |
| Residuals | 14 | 455.83 | 32.56 | | |


Normal Q-Q Plot
Sample Quantiles
Theoretical Quantiles
model$resid
model$fitted
c(A)
c(B)
c(C)
c(Block)

# What Have We Learned?

We discussed blocking in a $2^k$ factorial design in this chapter.

We also discussed complete and partial confounding in a $2^k$ factorial design.